

CPR IMCQ 11

Fall 2025

Physiology and Biochemistry

KEY file

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Physiology

1. A 34-year-old woman comes to the physician for a routine examination. She is concerned about her kidney function because her aunt has chronic kidney disease. Vital signs are within normal limits. Physical examination shows no abnormalities. Serum studies of creatinine and urea concentrations are within the reference ranges. Glomerular filtration rate is estimated using inulin and the results of this study are shown:

Plasma inulin concentration = 4 mg/dL,

Urine inulin concentration = 360 mg/dL,

Urine volume = 1.2 mL/min.

Which of the following values best represents glomerular filtration rate of this patient?

- A. 60 mL/min
- B. 88 mL/min
- C. 108 mL/min
- D. 122 mL/min
- E. 144 mL/min

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Urine volume = 1.2 mL/min.

Which of the following values best represents glomerular filtration rate of this patient?

SOM.MK.II.BPM1.3.CPR.2.PHYS.0331. Distinguish between the use of inulin and creatinine clearances as measures of the glomerular filtration rate.

A. 60 mL/min

B. 88 mL/min

✓ C. 108 mL/min

D. 122 mL/min

E. 144 mL/min

$$\begin{aligned}\text{GFR} &= \text{Creatinine clearance} \\ &= UV/P \\ &= 360 \times 1.2 / 4 = 108\end{aligned}$$

2. A 56-year-old man comes to the physician because of a 3-month history of cough without expectoration, right-sided chest pain, and a 5-kg (11-lb) weight loss. Vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show extremely elevated serum concentration of vasopressin but low serum osmolarity and sodium concentration. A CT scan of the chest shows a 2-cmx3.2-cm mass in the right apical lobe. Which of the following sets of alterations in the renal functions is most likely expected in this patient?

Option	Urine Volume	Urine Osmolarity	Free water clearance
A	↑	↑	Zero
B	↓	↑	Positive value
C	↓	↓	Negative value
D	↑	↓	Positive value
E	↓	↑	Negative value

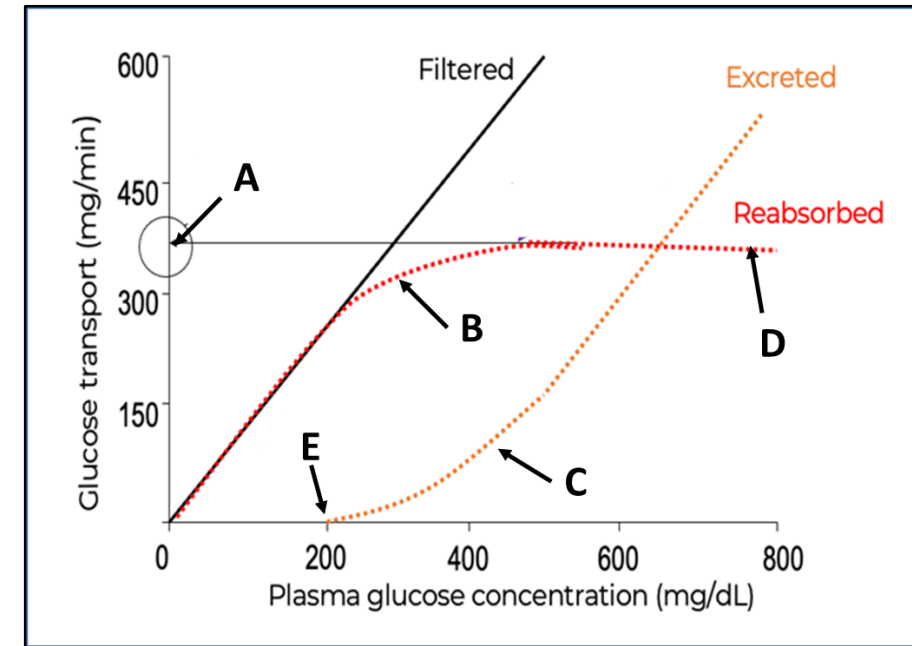
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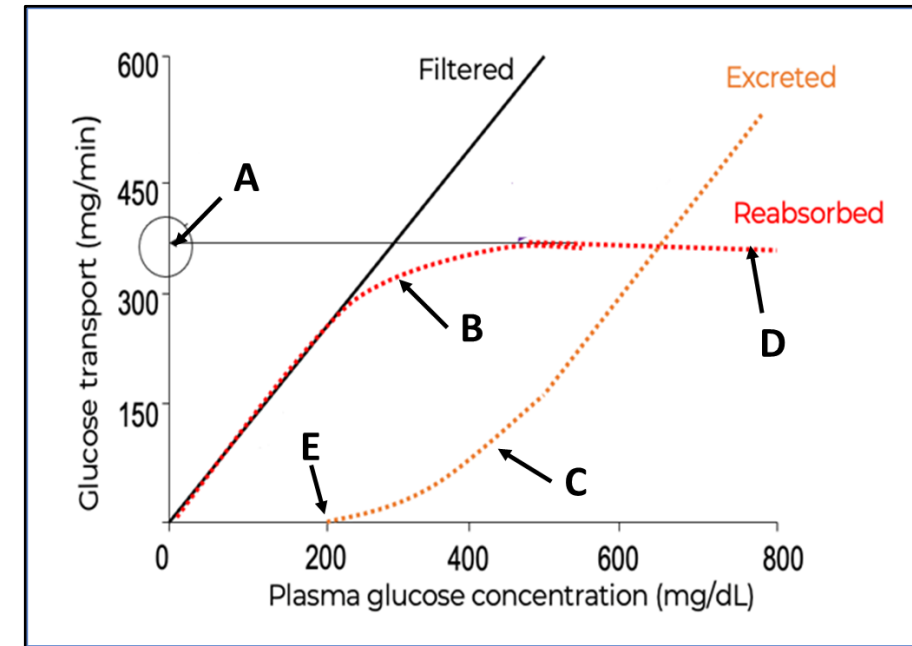
SOM.MK.II.BPM1.3.CPR.2.PHYS.0379. Describe the role of ADH in making the kidney to produce either a dilute or concentrated urine.

3. A 59-year-old man comes to the physician for a follow-up examination. He has a 9-year history of type 2 diabetes mellitus. He takes oral hypoglycemic drug and practices strict diet regimen. Vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show a random serum glucose concentration of 126 mg/dL (N: 140) and absence of urine glucose. Renal handling of glucose of the patient is shown in the graph. Which of the following mechanisms best explains label, B in the graph?



- A. All nephrons have the same transportation ability
- B. Nephrons have different transportation abilities (nephron heterogeneity)
- C. Serum glucose exceeds T_m glucose
- D. Serum glucose exceeds renal threshold
- E. Defective glucose reabsorption in PCT

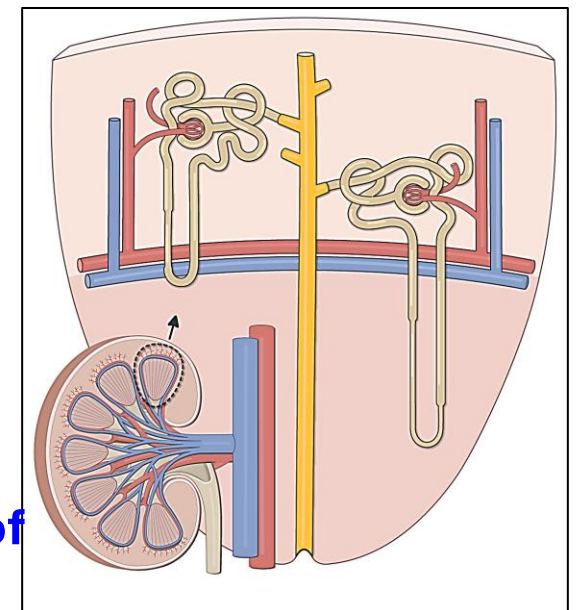
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SOM.MK.II.BPM1.3.CPR.2.PHYS.0381. With the help of a graph, explain tubular load, tubular transport maximum (T_{max}), renal threshold and splay for glucose.

Not all PCTs have the same number of transporters!!



4. A 26-year-old man participates in a study related to renal functions. He is intravenously infused with a very low concentration of para-amino hippuric acid (PAH). Results of laboratory studies are shown:

Urine volume = 1.0 mL/min, urine PAH = 10 mg/dL, plasma PAH = 0.02 mg/dL.
Hematocrit = 0.50.

Which of the following is the best estimate of the renal blood flow in this patient?

- A. 500 mL/min
- B. 700 mL/min
- C. 1000 mL/min
- D. 1200 mL/min
- E. 1400 mL/min

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Find out RPF:

= PAH Clearance =

$$UV/P = 10 \times 1/0.02 = 500.$$

Find out blood flow:

$$\begin{aligned} \text{RBF} &= \text{RPF}/(1 - \text{Hematocrit}) \\ &= 500/0.5 = 1000 \text{ mL/min.} \end{aligned}$$

SOM.MK.II.BPM1.3.CPR.2.PHYS.0330. Explain the clearance principle. Use the clearance equation and identify appropriate compounds used to estimate the glomerular filtration rate and renal plasma flow.

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12 Biochemistry Questions

A 29-year-old man comes to the physician because he is planning a high-altitude vacation. He has no history of major medical illness. His vital signs are within normal limits. He is prescribed acetazolamide to facilitate high-altitude acclimatization. Which of the following is most likely to be observed after administration of the medication?

- A. Increased urinary ammonium chloride
- B. Increased urinary acid phosphate
- C. Decreased tubular H^+ secretion
- D. Increased tubular bicarbonate reabsorption
- E. Metabolic alkalosis
- F. Low urine pH

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A 70-year-old man is admitted to the hospital because of a 12-hour history of difficulty in breathing. He has a 1-year history of congestive heart failure and is noncompliant with his medications. His pulse is 110/min, respirations are 26/min, and blood pressure is 130/90 mmHg. Physical examination shows deep, rapid, and labored breathing. He has bilateral pitting edema extending to his thighs. The laboratory studies shown are most likely the direct result of which of the following metabolic disturbances?

- A. pH: High; PCO₂: Low; HCO₃⁻: Low
- B. pH: Low; PCO₂: Low; HCO₃⁻: Low
- C. pH: Low; PCO₂: High; HCO₃⁻: Normal
- D. pH: High; PCO₂: Low; HCO₃⁻: High
- E. pH: Low; PCO₂: High; HCO₃⁻: High
- F. pH: High; PCO₂: Low; HCO₃⁻: Normal

Parameter	Patient	Reference
Sodium (Na+)	127 mEq/L	136-146 mEq/L
Potassium (K+)	5.2 mEq/L	3.5-5.0 mEq/L
Chloride (Cl-)	79 mEq/L	95-105 mEq/L
Creatinine	0.68 mg/dL	0.6-1.2 mg/dL
Glucose (Random)	120 mg/dL	<140 mg/dL
Lactate	25 mg/dL	4.5-19.8 mg/dL
Anion Gap	37	8-14 mEq/L

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A 45-year-old man is brought to the emergency department because of a 6-hour history of substernal chest pain radiating to his left jaw. He also has shortness of breath, nausea, and excessive sweating. His blood pressure is 165/90 mm Hg, pulse is 112/min, respirations are 20/min. Pulse oximetry on room air shows an oxygen saturation of 89%. Electrocardiogram shows ST segment elevation in leads I, aVL, V5 and V6. Results of laboratory studies are shown in the table. Which of the following pathophysiologic changes best explains the clinical presentation observed in this patient?

Laboratory	Laboratory Value	Normal value range
Troponin I	3.4 ng/mL	< 0.04 ng/mL
Hemoglobin	15 g/dL	13.5-17.5 g/dL
Total cholesterol	250 mg/dL	< 200 mg/dL
Serum myoglobin	250 ng/mL	25-72 ng/mL
Serum lactate	5.8 mmol/L	0.5 – 2.2 mmol/L

- A. Increased blood perfusion of the liver
- B. Increased NADH/NAD⁺ ratio in cardiac myocytes
- C. Increased activity of the TCA cycle in cardiac myocytes
- D. Low NADH/NAD⁺ ratio in skeletal muscles
- E. Upregulation of pyruvate dehydrogenase

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A 67-year-old man comes to the physician because of a 2-day history of progressive chest pain, cough, sweating, chills, and shortness of breath. His temperature is 39.4°C (103°F), pulse is 100/minute and regular. He is prescribed antibiotics. Synthesis of which of the following proteins is most likely increased in his liver?

- A. Albumin
- B. Transferrin
- C. Transcortin
- D. Immunoglobulin M
- E. C-reactive protein

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A 3-day-old newborn male is brought to the emergency department by his mother because episodes of floppiness, cool extremities, and breathing difficulty. He is also feeding poorly. He is afebrile, respirations are 35/min, pulse is 100/min, and blood pressure is within normal limits. Results of arterial blood gas studies show:

pH: 7.2 (7.35-7.45)

PO₂: 95 mmHg (75-105 mmHg)

PCO₂: 28 mm Hg (33-45 mm Hg)

Bicarbonate: 14 mEq/L (22-28 mEq/L)

Urinalysis shows presence of 3-hydroxy-3-methyl-glutaric acid and a diagnosis of HMG-CoA lyase deficiency is made. Which of the following best explains the acid-base disorder in this patient?

- A. Overproduction of non-volatile acids
- B. Increased volatile acid production
- C. Hypoventilation and reduced gas exchange
- D. Vomiting with volume contraction
- E. Respiratory alkalosis due to hyperventilation

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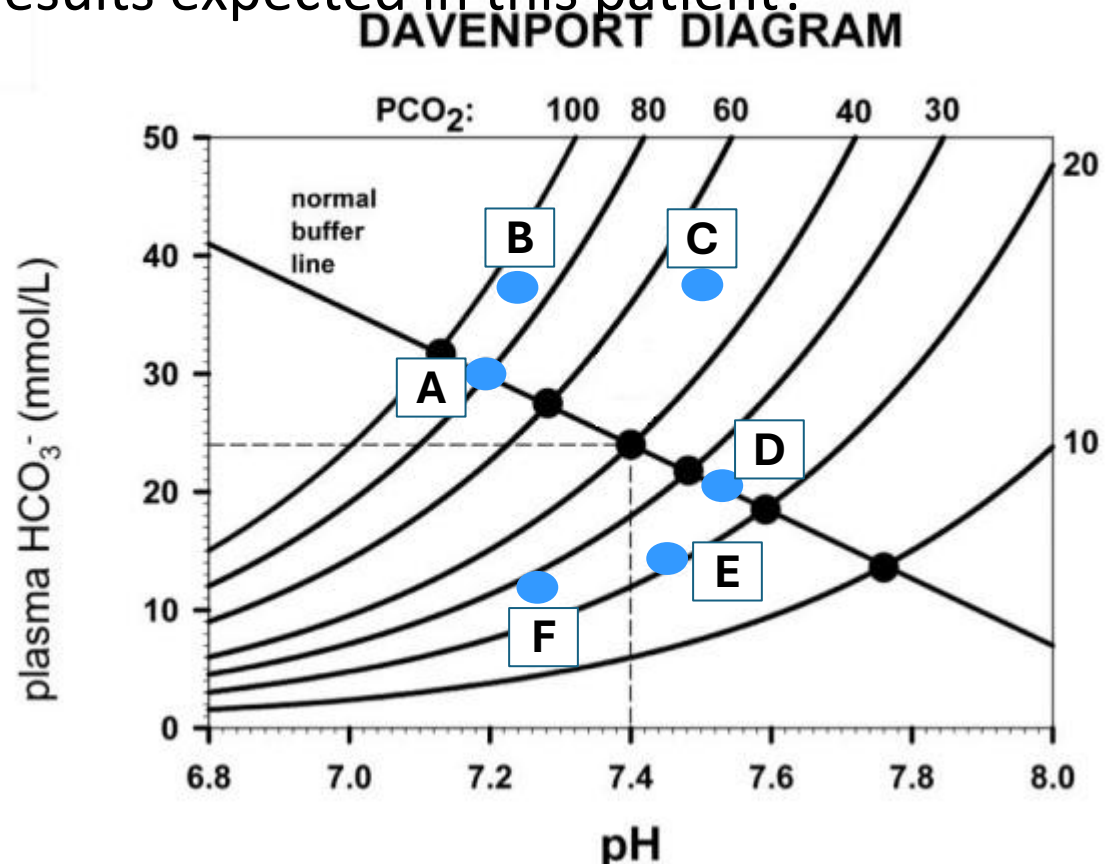
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A 6-week-old boy is brought to the emergency room because of poor feeding for the past 2 days. He has had vomiting from 3 weeks of age, initially non-bilious and in small amounts, which progressed to larger amounts after every meal. His mother has reduced breastfeeding and has supplemented with formula. His pulse is 180/min, blood pressure is 70/50mmHg, respirations are 10/min, and weight is 3.7 kg. Physical examination shows a listless infant with depressed fontanelles, dry mucus membranes, and abnormal skin turgor. His blood gas results are plotted on a Davenport diagram. Which of the following blue points best correlates with the blood gas results expected in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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A. A

B. B

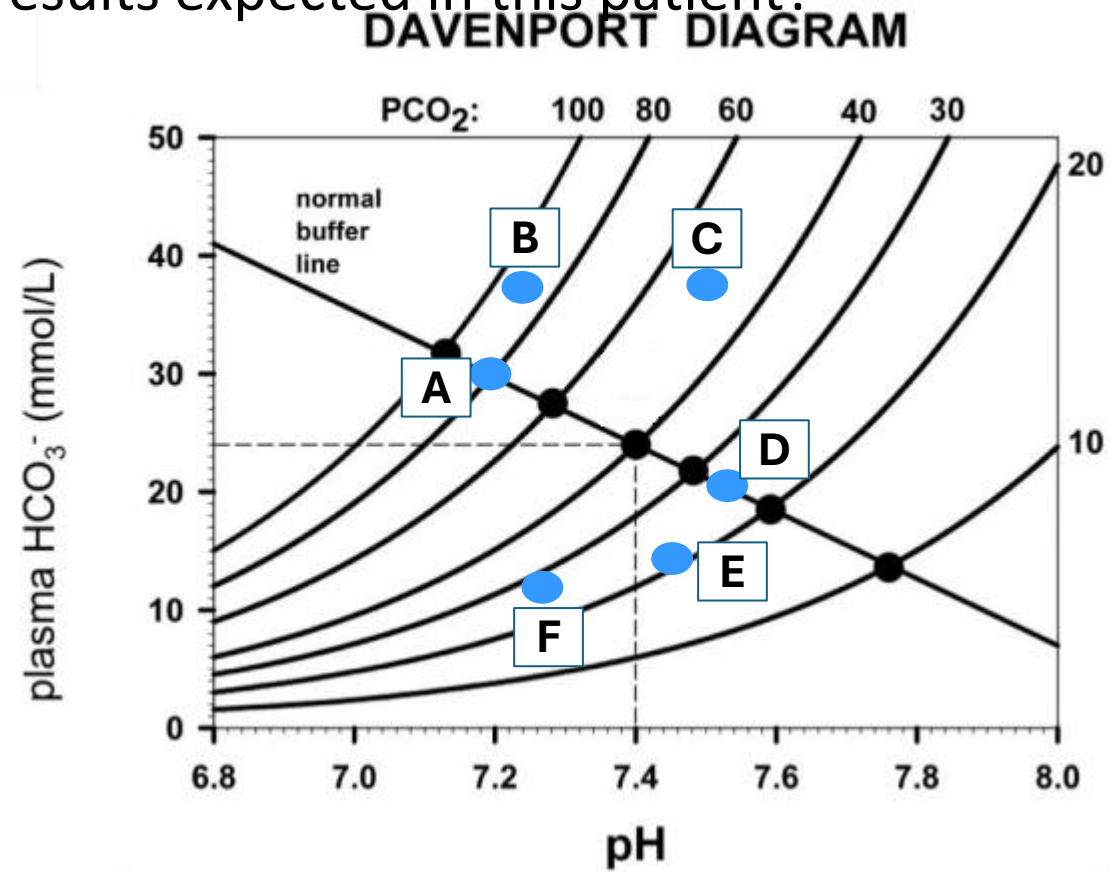
✓ C. C

D. D

E. E

F. F

SOM.MK.I.BPM1.3.CPR.1.INTG. 0025; Interpret ABG data (pH, PCO₂ and HCO₃⁻) and solve problems related to disorders of acid base balance using Davenport diagram. From blood values, identify simple and mixed metabolic and respiratory acid-base disturbances.



A 25-year-old woman is brought to the physician because of a 4-day history of difficulty speaking and abdominal pain. She describes the pain as constant, with no aggravating or relieving factors and 5/10 on the pain scale. Physical examination shows an anxious patient in mild distress. Her speech is slurred, mucus membranes icteric, and gait is slow with involuntary movements. Genetic testing confirms that she has autosomal recessive variants in the *ATP7B* gene. Which of the following laboratory findings is most likely expected in this patient?

- A. Elevated serum alpha 1-antitrypsin and alpha-fetoprotein levels
- B. Low serum ceruloplasmin levels
- C. Elevated serum ceruloplasmin levels
- D. Elevated serum hemopexin and haptoglobin levels
- E. Elevated serum transferrin levels

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- C. Elevated serum ceruloplasmin levels
- D. Elevated serum hemopexin and haptoglobin levels
- E. Elevated serum transferrin levels

A 15-year-old woman is brought to the emergency department by her parents because of a 2-day history of progressive bluish discoloration of her hands and feet. She has been on oral dapsone for the treatment of acne vulgaris for 3 weeks. She had a headache, dizziness, and myalgia in the previous week, with spontaneous improvement. Laboratory studies show a maximum oxygen saturation of 83% on room air, and MetHb level is 9.7% (reference: 0.0-1.5%). She is treated with 1mg/kg of methylene blue intravenously. Which of the following biochemical changes best explains her clinical presentation?

- A. Glycosylation of hemoglobin
- B. Inhibition of hexokinase
- C. Decreased blood pH
- D. Increased binding of 2,3-BPG to hemoglobin
- E. Oxidation of ferric heme
- F. Oxidation of ferrous iron in heme
- G. Reduction of ferric iron

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A 9-month-old girl is brought to the emergency department by her mother because of cough and breathing difficulty for the past 30 minutes. She has had multiple episodes of vomiting in the preceding 6 hours. Her pulse is 205/min, respirations are 30/min and temperature is 38.5°C. Physical examination shows deep and labored breathing, and bilateral crackles on auscultation. Laboratory studies show plasma salicylate levels are 76 mg/dL (normal: <20 mg/dL). Her arterial blood gases are measured. Which of the following blood gas parameters is most likely in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI

	pH (7.35-7.45)	HCO₃⁻ (22-28 mEq/L)	PaCO₂ (33-45 mmHg)
I	7.25	26	55
II	7.35	15	60
III	7.5	25	20
IV	7.55	46	55
V	7.35	40	65
VI	7.46	48	20

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SOM.MK.I.BPM1.3.CPR.1.INTG.0017; Describe processes that lead to respiratory acid-base disturbances (acidosis and alkalosis) and identify common causes for each.

	pH (7.35-7.45)	HCO ₃ ⁻ (22-28 mEq/L)	PaCO ₂ (33-45 mmHg)
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III	7.5	25	20
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VI	7.46	48	20

A 24-year-old man is brought to the emergency department because of a 6-hour history headache, nausea and dizziness. He has traveled from New York to Aspen, Colorado (7900 ft above sea level). Arterial blood gas analysis on admission show pH=7.5 (7.4); $\text{PCO}_2 = 30$ (40 mm Hg); $\text{HCO}_3^- = 23$ (22-24 mEq/L). Which of the following is most likely expected one week later?

- A. Increased urinary ammonium excretion
- B. Increased erythrocyte 2,3-bisphosphoglycerate levels
- C. Left shift of hemoglobin oxygen dissociation curve
- D. Inhibition of renal erythropoietin secretion
- E. Increased tubular H^+ secretion
- F. Increased hemoglobin oxygen affinity

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SOM.MK.I.BPM1.3.CPR.1.INTG. 0024; Distinguish between acute and chronic respiratory disturbances. Describe the causes, compensatory responses for: Acute and Chronic Respiratory acidosis – Causes of hypoventilation. Acute and Chronic Respiratory alkalosis – Causes of hyperventilation, high altitude adaptations.

- A. Increased urinary ammonium excretion
- ✓ B. Increased erythrocyte 2,3-bisphosphoglycerate levels
- C. Left shift of hemoglobin oxygen dissociation curve
- D. Inhibition of renal erythropoietin secretion
- E. Increased tubular H^+ secretion
- F. Increased hemoglobin oxygen affinity

A 23-year-old woman is brought to the emergency room because of a 3-hour history of difficult breathing. She has a 5-day history of fever with productive cough. Her respirations are 36/minute and shallow. Laboratory studies show arterial pH is 7.26 (7.35-7.45), PaCO_2 is 75 mmHg (33-45) and serum bicarbonate level is 25 mEq/L (22-28). Which of the following is most likely to be observed after 3 days in this patient?

- A. Stimulation of respiratory center to increase respiratory rate
- B. Increased proton secretion by the renal tubular cells
- C. Loss of bicarbonate in urine and a decrease in serum bicarbonate levels
- D. Inhibition of renal carbonic anhydrase to lower serum bicarbonate levels
- E. A decrease in respiratory rate by inhibition of the respiratory center

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SOM.MK.I.BPM1.3.CPR.1.INTG. 0017; Describe processes that lead to respiratory acid-base disturbances (acidosis and alkalosis) and identify common causes for each.

SOM.MK.I.BPM1.3.CPR.1.INTG. 0018; Describe processes that lead to metabolic acid-base disturbances (acidosis and alkalosis). Identify common causes for each including (diabetic ketoacidosis, lactic acidosis, diarrhea, acetazolamide treatment and Vomiting, pyloric stenosis).

A 45-year-old woman comes to the physician for a follow-up examination because she had abnormal laboratory reports. Her previous laboratory results show: Serum cholesterol is 310 mg/dL (<200 mg/dL), LDL-cholesterol is 280 mg/dL (<160 mg/dL), and triglycerides is within reference limits. Physical examination shows no abnormalities. Which of the following best describes the findings on serum protein electrophoresis?

- A. Increased gamma globulins
- B. Decreased albumin and increase in alpha-1 globulins
- C. Increased β -globulins
- D. Decreased alpha-2 globulins with increased β -globulins
- E. Increased albumin levels

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